

Dual-EMA Trend (BTC + ETH)

A standardized empirical study — crypto

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Abstract

This study documents the historical risk-adjusted performance of the Dual-EMA Trend (BTC + ETH) strategy over the period 2017-01-02 to 2026-06-07. Across 3,444 trading days the strategy realised an annualised Sharpe ratio of 1.12, a compound annual growth rate of 39.14%, and a maximum drawdown of -36.77%. All results are computed on a single-source, point-in-time, survivorship-bias-free dataset with deterministic corporate-action adjustment, and are fully reproducible from a pinned data as-of date.

Keywords: *crypto, backtest, risk-adjusted returns*

Disclaimer: This document is for informational and educational purposes only and does not constitute investment advice. Past performance, including backtested results, is not indicative of future results.

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1. Introduction

1.1 Motivation

We evaluate Dual-EMA Trend (BTC + ETH) as a candidate for the strategy book. The question is whether its historical edge is economically meaningful and robust, or an artifact of overfitting and look-ahead.

1.2 Preview of findings

Over the sample the strategy earns a Sharpe ratio of 1.12 (Sortino 1.72, Calmar 1.06) with annualised volatility of 34.78%. _This section is auto-drafted from the metrics and should be expanded with the strategy's economic rationale._

2. Data and Methodology

2.1 Data

Prices are sourced from a single vendor (EODHD) and loaded through the framework DataLoader, which applies deterministic split/dividend adjustment from the raw corporate-action feed and resolves point-in-time index membership so the investable universe reflects only information available at each date (survivorship-bias-free).

2.2 Backtest methodology

Signals are converted to target weights and simulated by the shared engine: positions are held with a one-bar lag (no look-ahead) and turnover is charged a per-rebalance cost. Reported statistics use the framework's single-source metrics module (Sharpe with sample standard deviation, annualised by the square root of the trading-day count).

$$\text{Sharpe} = \frac{\bar{r}}{\sigma_r} \sqrt{P}$$

3. Results and Analysis

Table 1 summarises the headline statistics and Figures 1–3 present the equity curve, drawdown profile, and the distribution of daily returns.

The strategy compounds at 39.14% per annum with a worst peak-to-trough decline of -36.77%, implying a Calmar ratio of 1.06. The realised win rate is 52.52%. _Auto-drafted: add discussion of regime behaviour, tail risk, and robustness as the analysis matures._

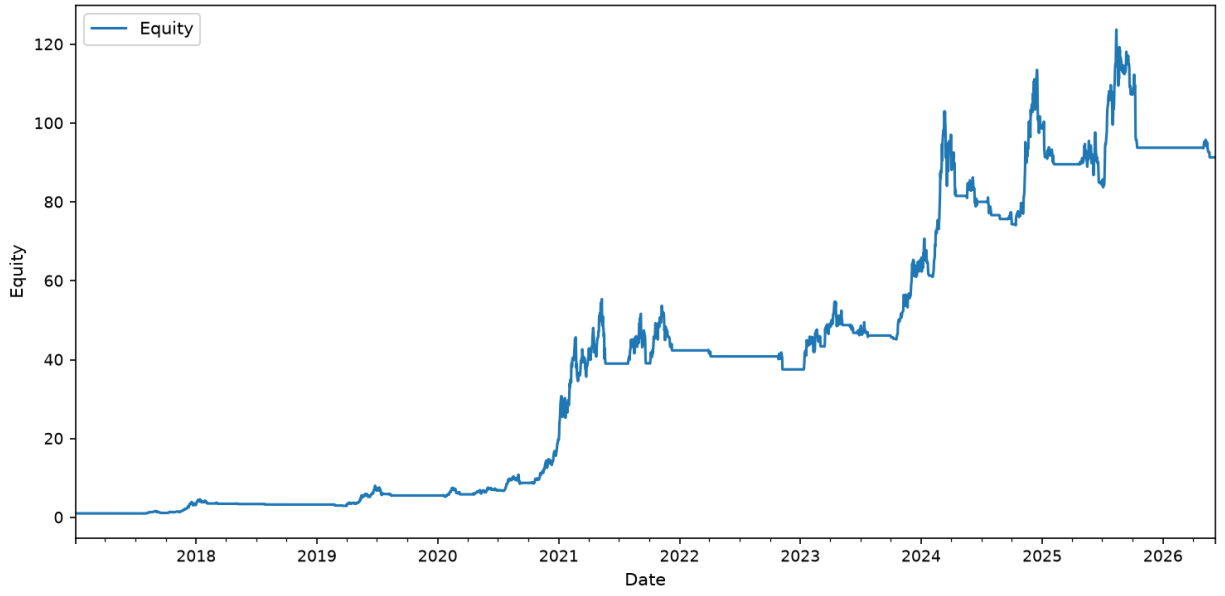


Figure 1: Cumulative equity curve (growth of \$1 invested).

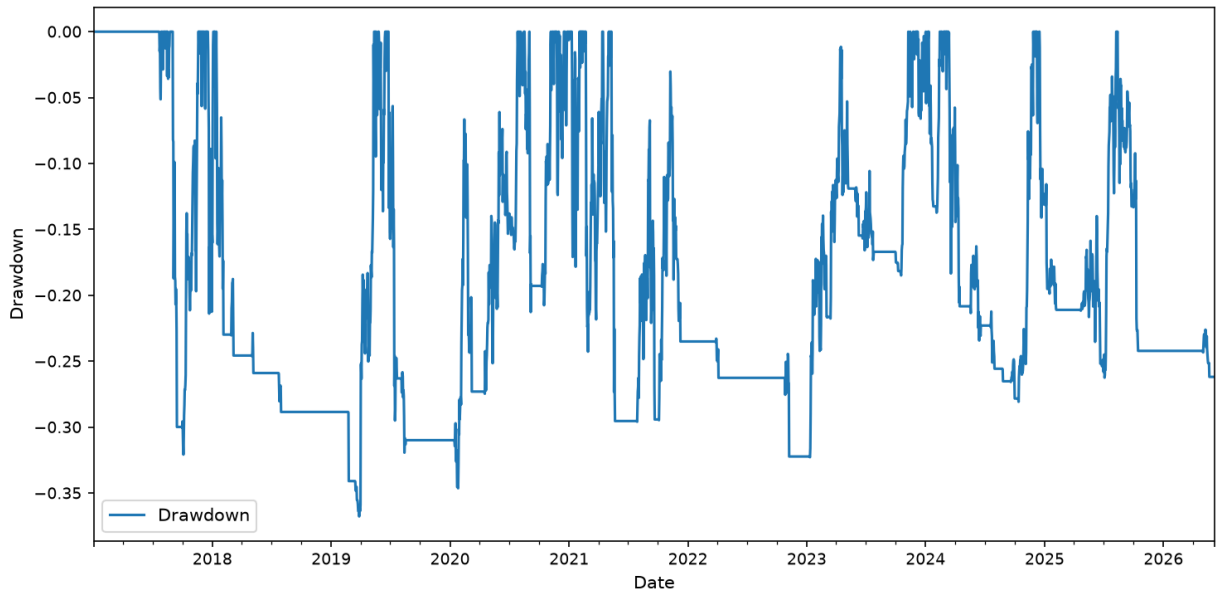


Figure 2: Drawdown (underwater) curve.

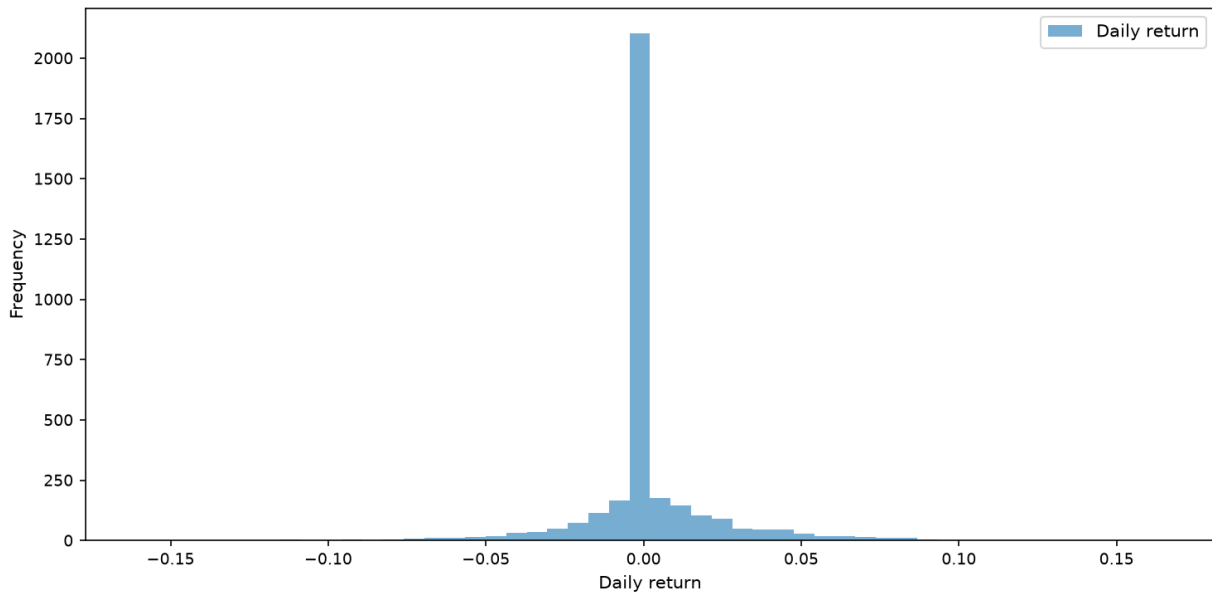


Figure 3: Distribution of daily returns.

Table 1: Performance summary statistics over the full backtest period.

Metric	Value
Period	2017-01-02 to 2026-06-07
Observations	3,444
CAGR	39.14%
Annualised volatility	34.78%
Sharpe ratio	1.12
Sortino ratio	1.72
Calmar ratio	1.06
Maximum drawdown	-36.77%
Total return	9032.55%
Win rate	52.52%

4. Conclusion

The Dual-EMA Trend (BTC + ETH) strategy delivers a historically attractive, reproducible risk-adjusted return on a survivorship-bias-free sample. Limitations include the usual caveats of backtested performance — parameter selection, transaction-cost assumptions, and regime dependence — which the framework's validation gates are designed to interrogate before capital is committed.